Experimental Determination of Moment of Inertia

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Contents

1 Abstract

1.1

This report investigates the moment of inertia of rotating bodies using experimental measurements of angular velocity as a function of time. Linear regression was used to determine the angular acceleration from the measured data, and the results were compared for different experimental configurations. The aim was to examine how the distribution of mass affects the moment of inertia and to evaluate how well the experimental results agree with theoretical expectations.

2 Introduction

2.1

Moment of inertia is a physical property of rotating bodies and describes their resistance to changes in angular velocity. Since it affects how much torque is required to accelerate or decelerate an object, knowing how to calculate and measure it is important in many mechanical applications. In this study, several practical methods for measuring moment of inertia were explored and compared with theoretical values.

3 Theory/Background

3.1

The forces involved are

$$\sum F = mg - T = ma$$

a = the linear acceleration of the falling mass
tourqe are given by T times the distance to the perpendicular line the force are aplied on

$$Tr = \tau$$

Newton second law for analogus with the law for translation τ are substituting F and I are substituting the mass of the object adding the spacial distribution of the object to the equation.

$$\tau = I\alpha$$

rotation

$$I = \frac{mgr}{\alpha} + mr^2$$

4 Method

4.1

5 Results

5.1

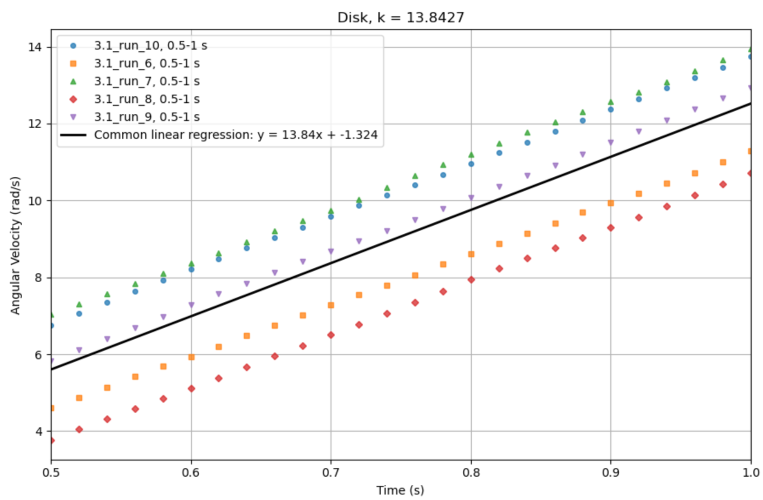

Figure 1: Graph from the experiment used to determine the moment of inertia of a disk. The k-value gives the angular acceleration.

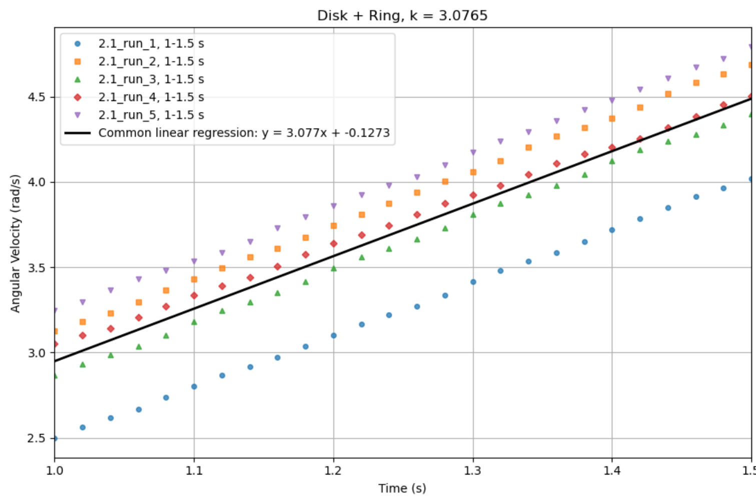

Figure 2: Graph from the experiment used to determine the moment of inertia of a disk with added weight. The k-value gives the angular acceleration.

6 Discussion

6.1

7 References

7.1

8 Appendix

8.1

9 Contributions

9.1

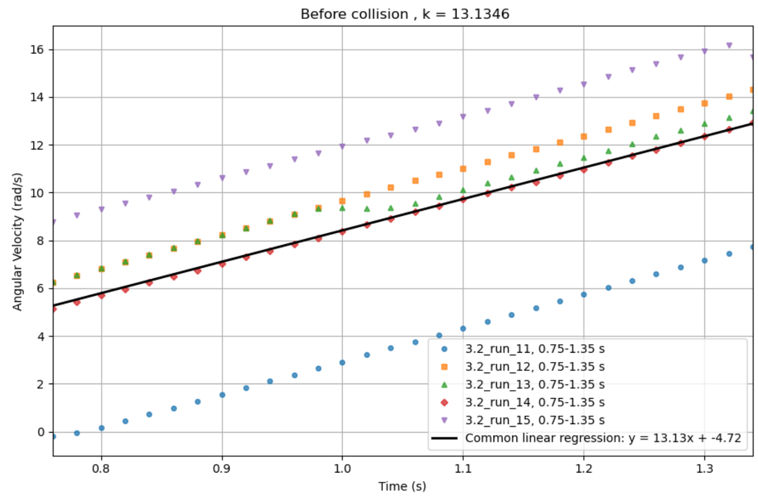

Figure 3: Graph showing the angular speed before impact of the added weight. The k-value gives the angular acceleration.

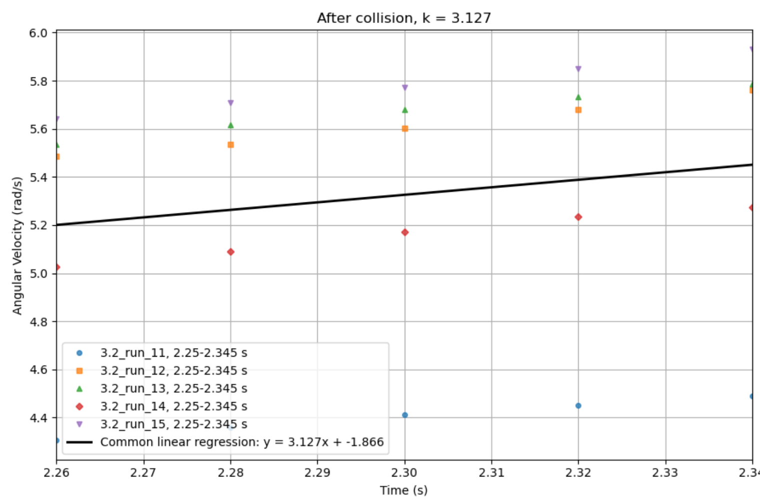

Figure 4: Graph showing the angular speed after impact of the added weight. The k-value gives the angular acceleration.